

Practice Problems for PQ10



For **PQ10 - Inheritance and Overriding** you'll implement one class using inheritance. This exam covers Chapters 8 (classes) and the first part of Chapter 9 (inheritance) in your textbook.

For this exam you'll need to know how to:

- Read a UML diagram and create a class that matches the diagram
- Define instance variables, accessors and mutators
- Define constructors, both for stand-alone and subclasses
- Override the `toString` method in a subclass. (Make sure you also know how to use `String.format()` correctly from Chapter 4).

As on the previous exams, your class will be tested for syntax correctness (name, public, return type and parameters) and for correctness. Work step-by-step so that you get partial credit. You will find some inheritance problems on Practicelt (in Chapter 9).

The problems shown here are samples. The problem you will receive will be similar to, but not the same as one of these problems.

1 AUTO

Complete the `Auto` class below so that it inherits from the given `Vehicle` class.

2 CAR

Complete the Car class below so that it inherits from the given Vehicle class.

3 QUARTER

Form a subclass Quarter from the class Coin. A quarter has a state theme.

4 TITLED RECTANGLE

A Rectangle whose title is centered inside itself

5 FILLED RECTANGLE

A Rectangle whose interior is filled with a pattern.

6 LABELEDPOINT TOSTRING

Override the toString method in the LabeledPoint class

7 LABELEDPOINT EQUALS

Override the equals method in the LabeledPoint class

8 AUTO EQUALS

Override the equals method in the Auto class

9 SAVINGSACCOUNT CONSTRUCTOR

Define a constructor for the SavingsAccount subclass shown here.

10 SAVINGSACCOUNT ADDINTEREST

Enhance the addInterest method of the SavingsAccount class.

11 TIMEDEPOSITACCOUNT

Extends the BankAccount hierarchy with a TimedDepositAccount class.

12 NUMERICQUESTION

Add a class `NumericQuestion` to the question hierarchy of your text.

13 FILLINTHEBLANKQUESTION

Add a class `FillInTheBlankQuestion` to the question hierarchy of your text.

14 MULTICHOICEQUESTION

Add a class `MultiChoiceQuestion` to the question hierarchy of your text.

15 SQUARE

Define a class named `Square` that extends `Rectangle`.